

## Solution Requirements

|                      |                                              |
|----------------------|----------------------------------------------|
| <b>Date</b>          | 03 JULY 2026                                 |
| <b>Team ID</b>       | XXXXXX                                       |
| <b>Project Name</b>  | <b>A Comprehensive Measure of Well-Being</b> |
| <b>Maximum marks</b> | 4 Marks                                      |

### Define Functional and Non-Functional Requirements:

**Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).

**Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category   | Requirement Description                                                                                                                                                                                             | Priority |
|------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1    | Authentication         | Users can securely access the HDI Insight application to use the prediction and analysis features.                                                                                                                  | High     |
| 2    | Authorization Levels   | The system provides role-based access, allowing administrators to manage datasets and models while users can perform HDI predictions and view results.                                                              | Medium   |
| 3    | External Interfaces    | The application integrates with the HDI dataset, trained Machine Learning model, Flask backend, and visualization libraries to generate predictions.                                                                | High     |
| 4    | Transaction Processing | The system accepts user input (Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita), validates it, processes it through the ML model, and returns the predicted HDI category. | High     |
| 5    | Reporting              | The system displays prediction results, development insights, graphical visualizations, and allows users to review country-wise HDI information.                                                                    | High     |

|   |                                   |                                                                                                                                                                                        |        |
|---|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 6 | Business Rules                    | The prediction is generated only after validating all required inputs. The system classifies countries into Very High, High, Medium, or Low HDI categories based on the trained model. | High   |
| 7 | Compliance to Laws or Regulations | The application handles publicly available HDI data responsibly and follows standard data privacy and ethical AI practices.                                                            | Medium |
| 8 | Other                             | The system provides a user-friendly dashboard with interactive charts and development indicators for better understanding.                                                             | Medium |

## Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category               | Requirement Description                                                                                | Target Metric / Acceptance Criteria                 |
|------|----------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 1    | Performance & Speed        | The application should generate HDI predictions quickly after user input.                              | Prediction response time less than <b>2 seconds</b> |
| 2    | Scalability                | The system should efficiently support multiple users and future dataset updates                        | Support at least <b>500 concurrent users</b>        |
| 3    | Security & Data Privacy    | Protect user inputs and prevent unauthorized access to application resources.                          | Secure HTTP communication and input validation      |
| 4    | Reliability & Availability | The application should remain available with minimal downtime.                                         | 99% uptime                                          |
| 5    | Usability & Accessibility  | Provide a simple, responsive, and intuitive interface that works across devices.                       | Responsive UI compatible with major browsers        |
| 6    | Other                      | The application should be maintainable, modular, and easy to upgrade with new machine learning models. | Modular code structures and documentation           |